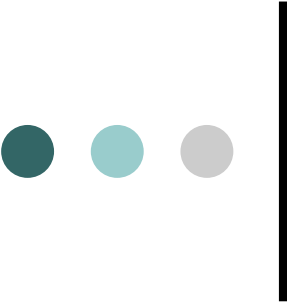




THE FLUID COMPARTMENTS

Dr ABDELOUAHAB



Plan

Definition of the internal environment.

Use of tracers in the study of liquid compartments.
Definition of a compartment.

.Measurement of compartment volume.

.Water in the body.

The major compartments of the body.

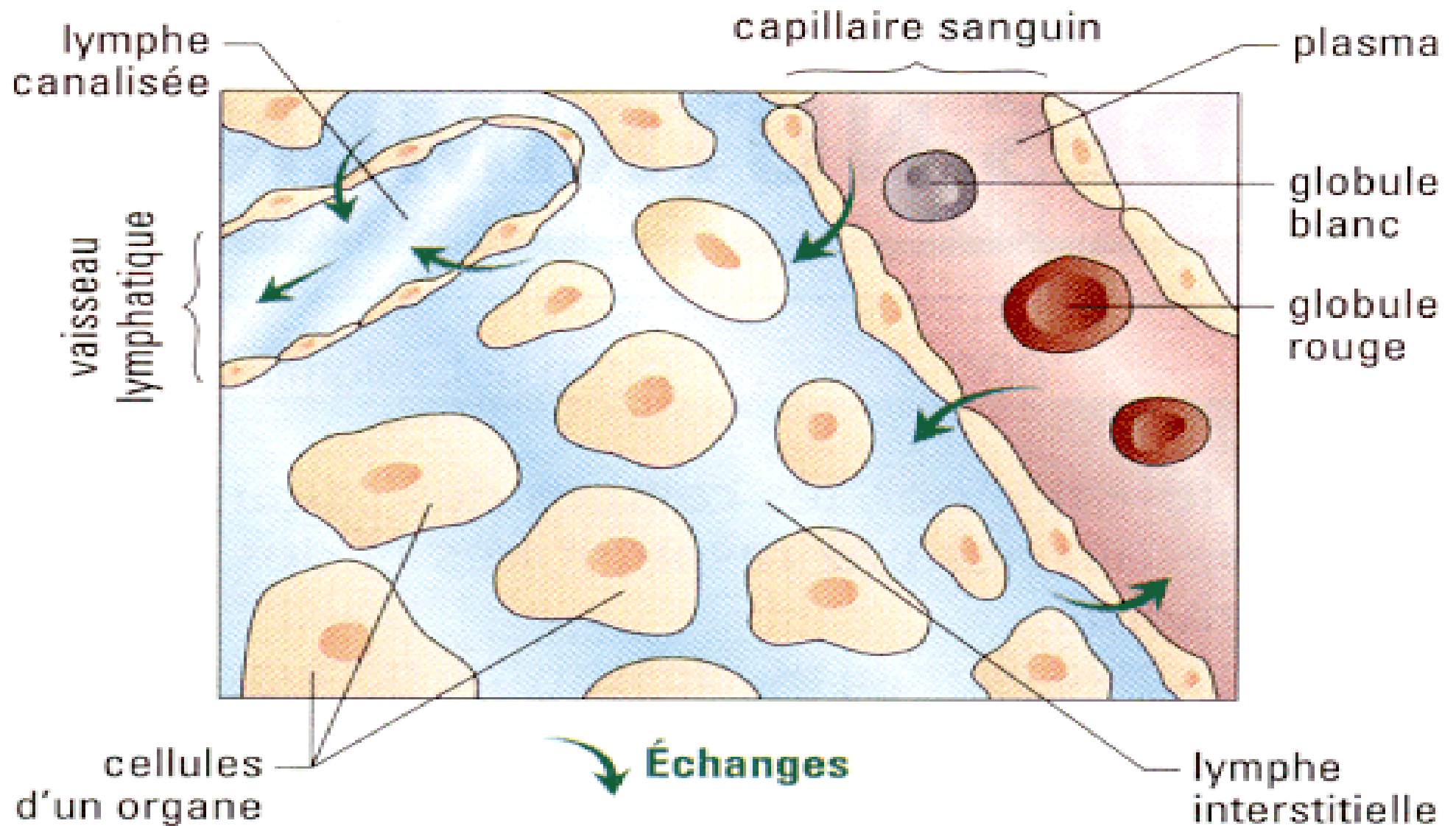
Extracellular

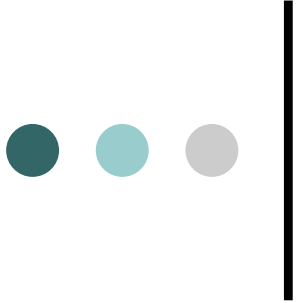
Intracellular

Transcellular

Exchanges between compartments.

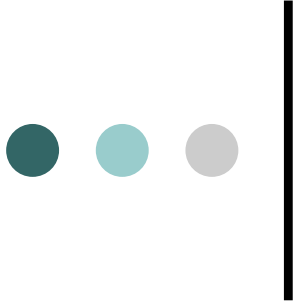
The internal environment





Definition internal environment

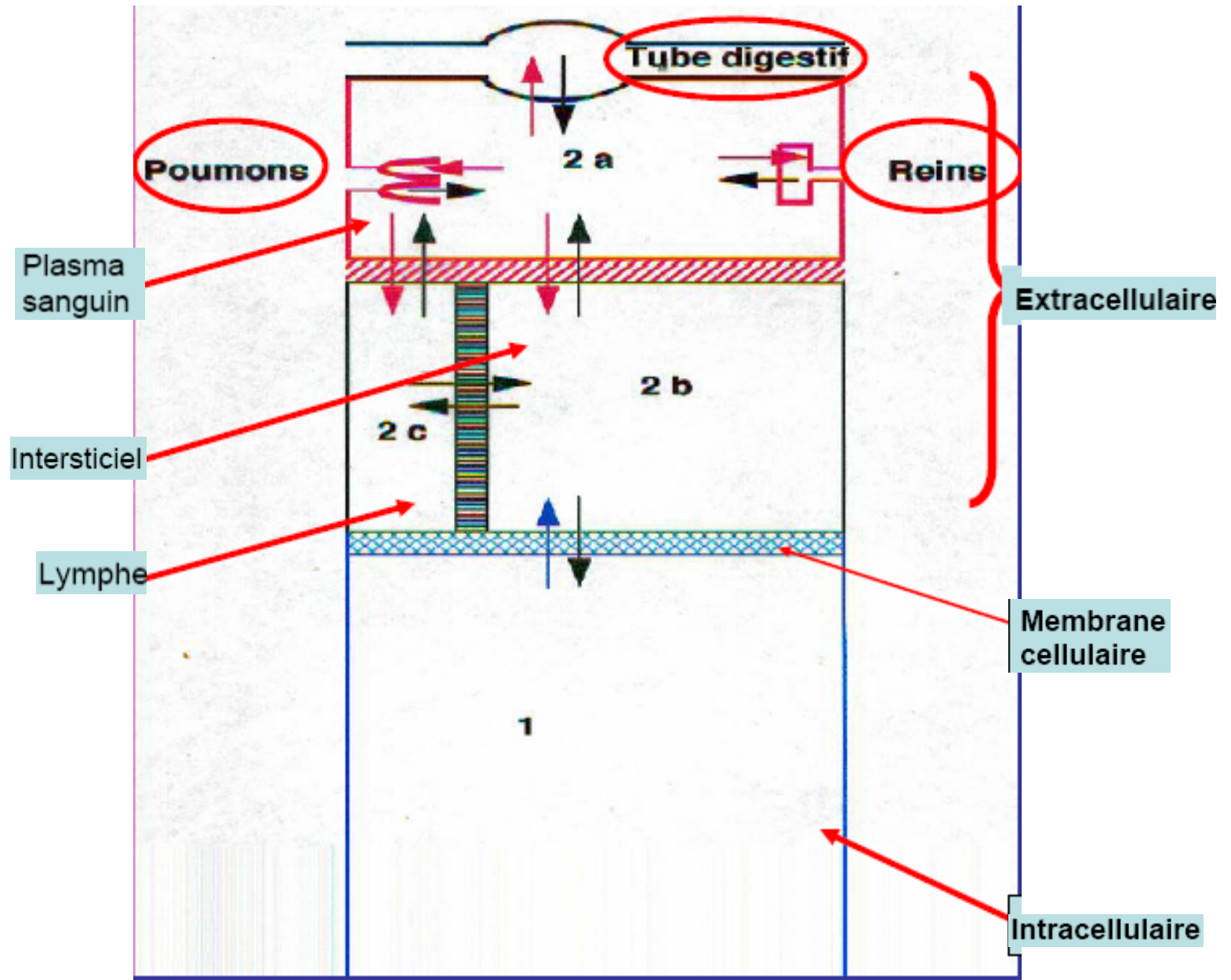
- The internal environment of a multicellular organism, such as humans, corresponds to the **extracellular fluid** contained in the body and separated from the external environment by the **skin tissue**.
- The extracellular fluid = **interstitial fluid + plasma**
- The extracellular fluid is renewed by two circulations : **blood circulation** and **lymphatic circulation**
- The **internal environment** is characterized by its



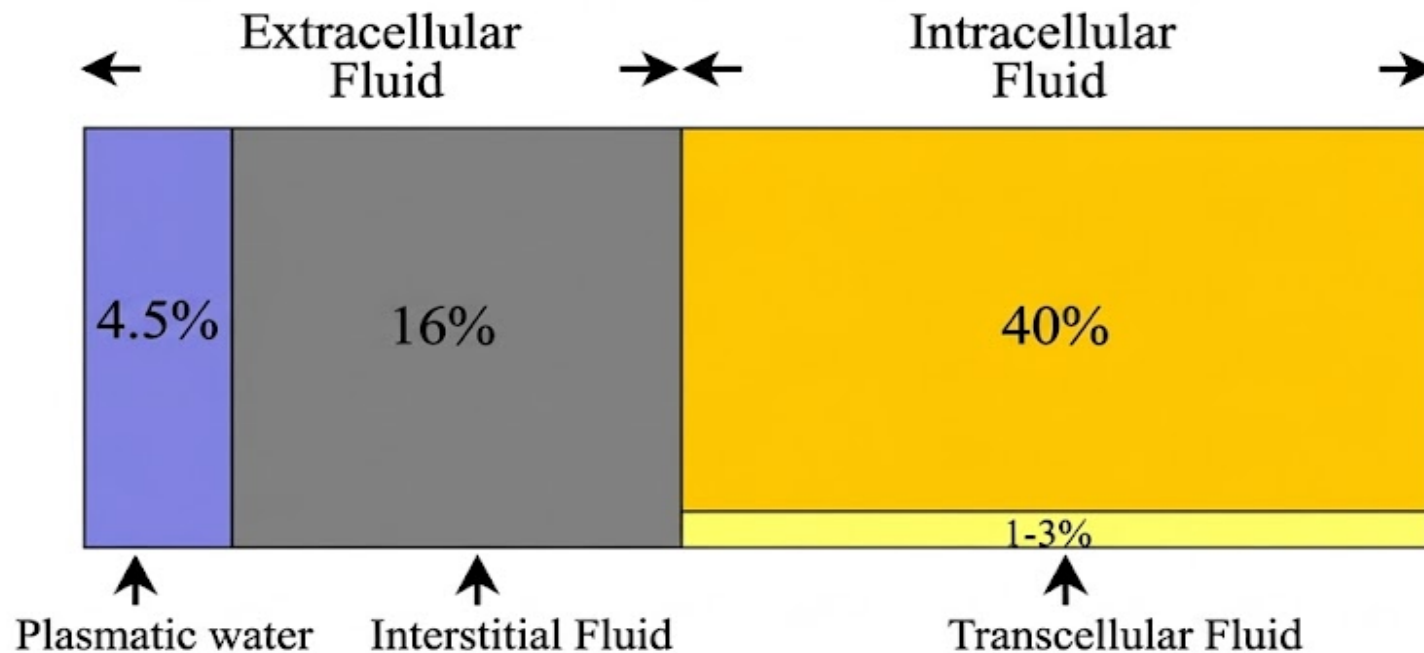
Homeostasis

- Imbalance → **receptor** → **regulatory systems** → correction of the imbalance
- Disorders of internal environment regulation are detectable through a **blood test**.

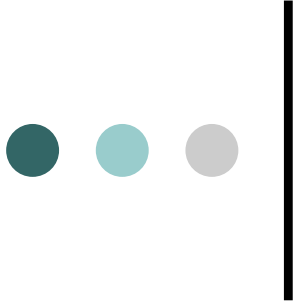
Fluid compartments of the body



Fluid compartments of the body

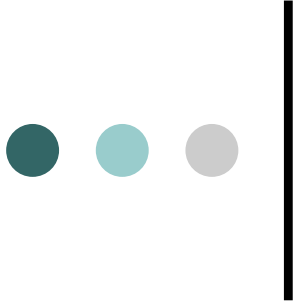


- 1- Different composition of dissolved substances in each compartment.
- 2- Access: Plasma water.
- 3- IC / EC exchanges across the cell membrane.
- 4- Plasma / interstitium exchanges = capillary endothelium.
- 5- Heterogeneity of the cellular compartment.



Fluid compartments of the body

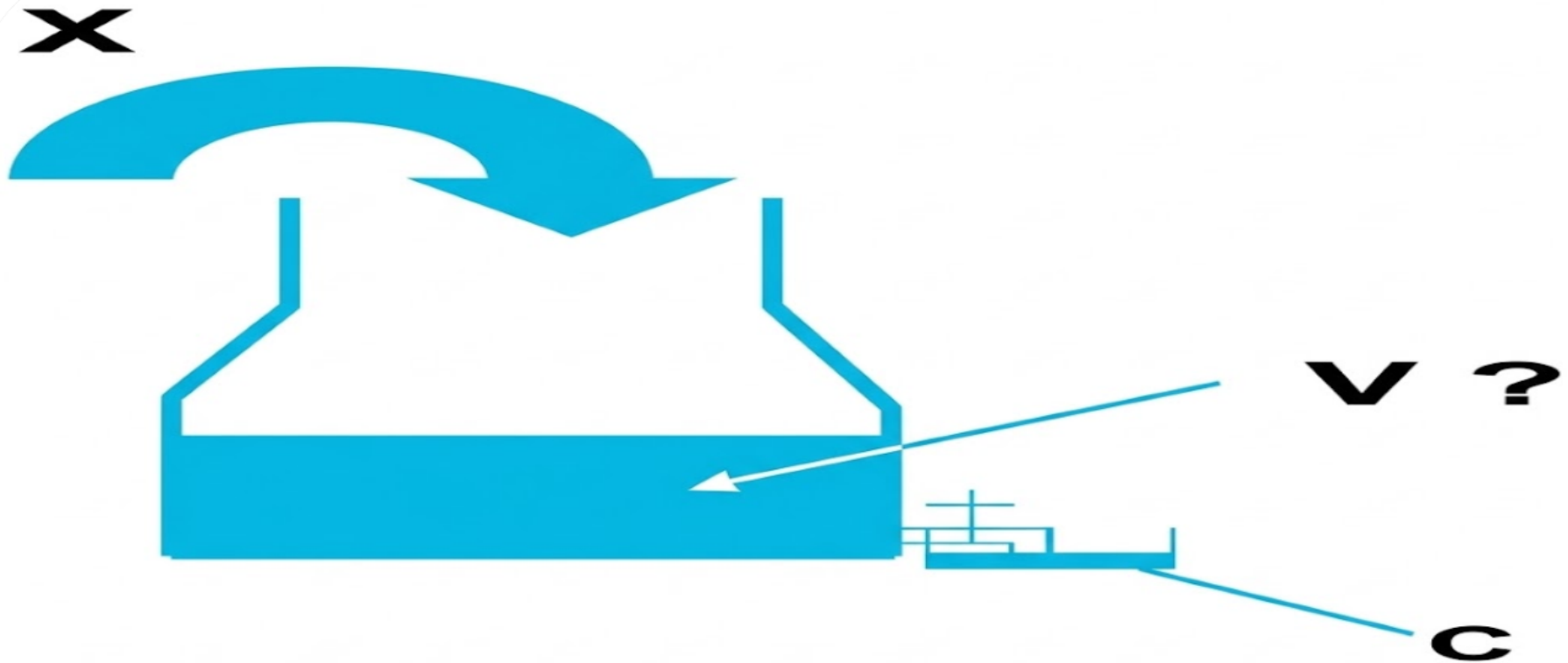
- **Tracers:**
- Non-toxic
- Injectable
- Homogeneous and specific diffusion
- No effect on water distribution
- Simple measurement



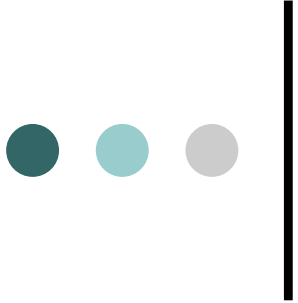
Fluid Compartments

- Measurement of fluid compartments: $V = M / C$
- M = mass of the injected indicator
- C = concentration in the sample
- Conditions to be respected in vivo: the tracer must be **non-toxic, easily measurable, with rapid and uniform distribution.**

Measurement of Fluid Compartments

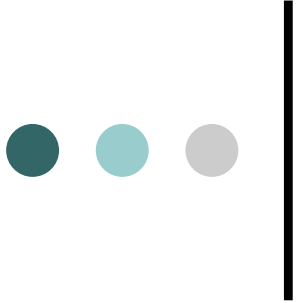


$$V = X / C$$



Results

- **Total body water:** tritiated water
Plasma volume: Evans blue, ^{125}I -labeled albumin, ^{51}Cr -labeled red blood cells
- **Extracellular water:** Inulin (20%)
Interstitial water: extracellular water – plasma water I
- **Intracellular water:** total body water – extracellular water



Total Water Homeostasis

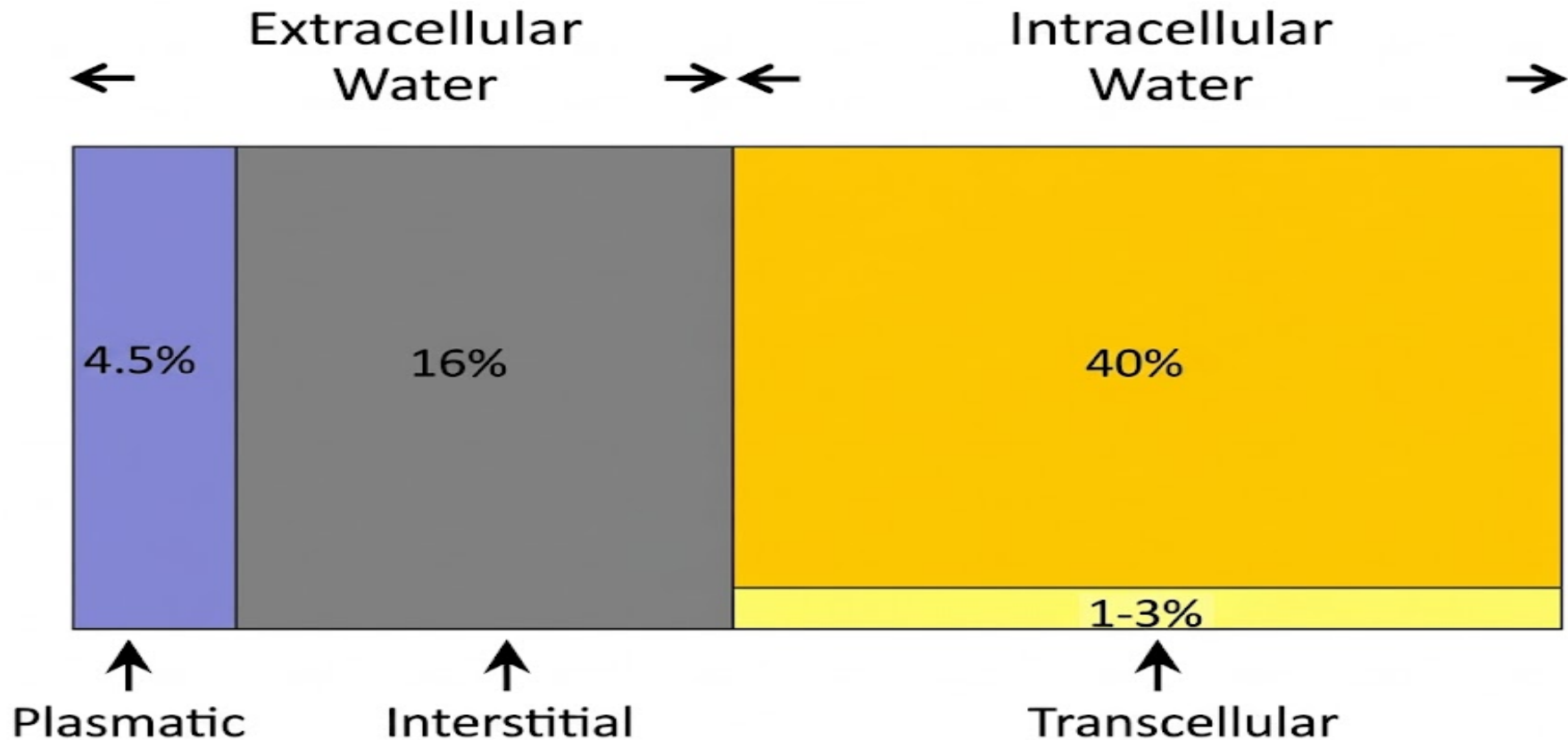
- **Beverages** → main source of water (1500 ml)
- **Food** → water contained in food (700 ml)
- **Endogenous water** → water produced by cellular metabolism (300 ml)

sorties	
urine	1500
Stool	100
sweating	0
perspiration	500
respiration	400
Total	2500

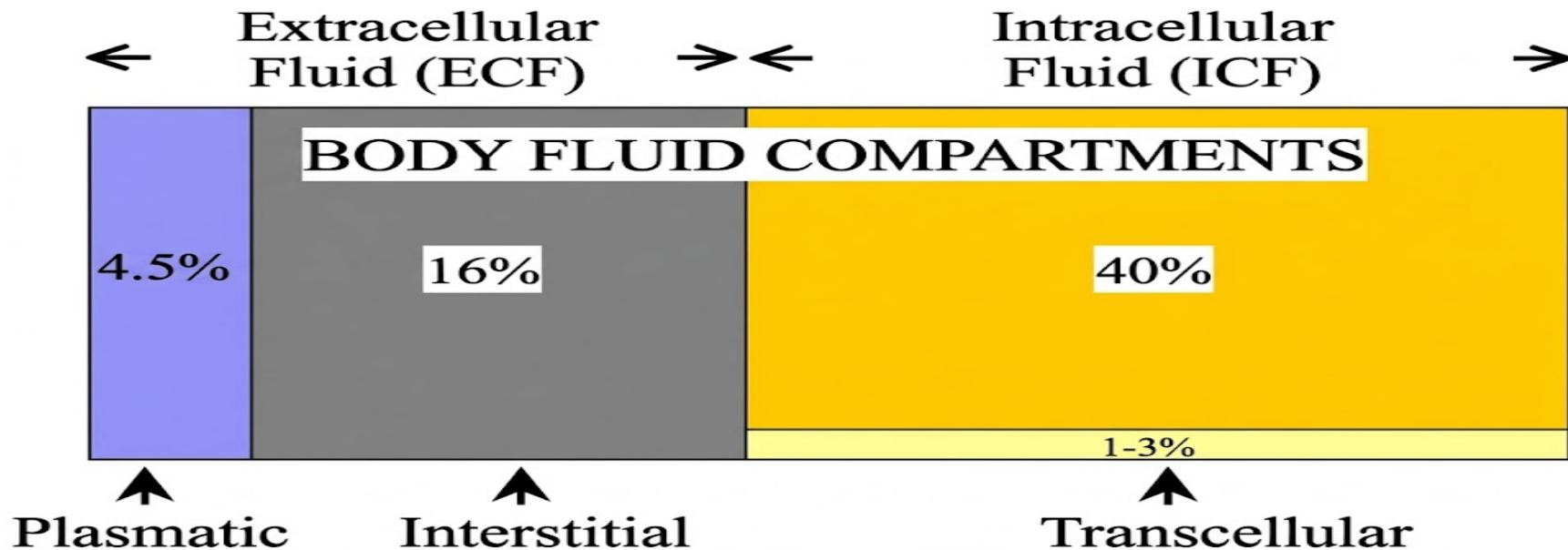
Regulation: thirst and kidney

Distribution of Water in the Body

Total body water = 60% of body weight



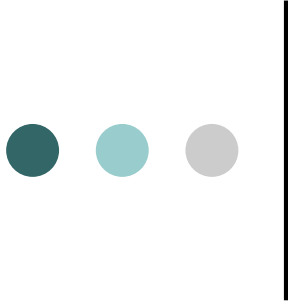
Distribution of Water in the Body



Total body water = 60% of body weight

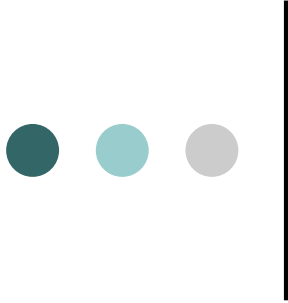
- Varies with tissue type: 70% in soft tissues, lower in bone, cartilage and fat (overweight)
- Varies with age: 75% in infants, less than 50% in elderly
- Varies with sex: 60% in males, 50% in females (pregnancy)

Any rapid change in body weight reflects water retention or water loss.



Results of the measurements

- In % of body weight | Total body volume | Heavy water or tritiated water
- **60%** of body weight
- Varies with **sex**
- Varies with **age**



Extracellular Compartments

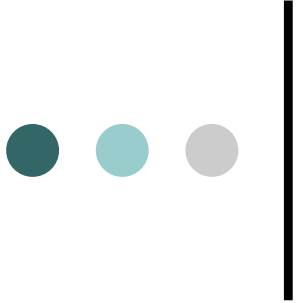
Total Volume

- Mannitol, -inulin C {14}
- 20% of body weight

Plasma Volume

- labeled Albumin
- Evans Blue dye
- 4% of body weight

Blood Volume

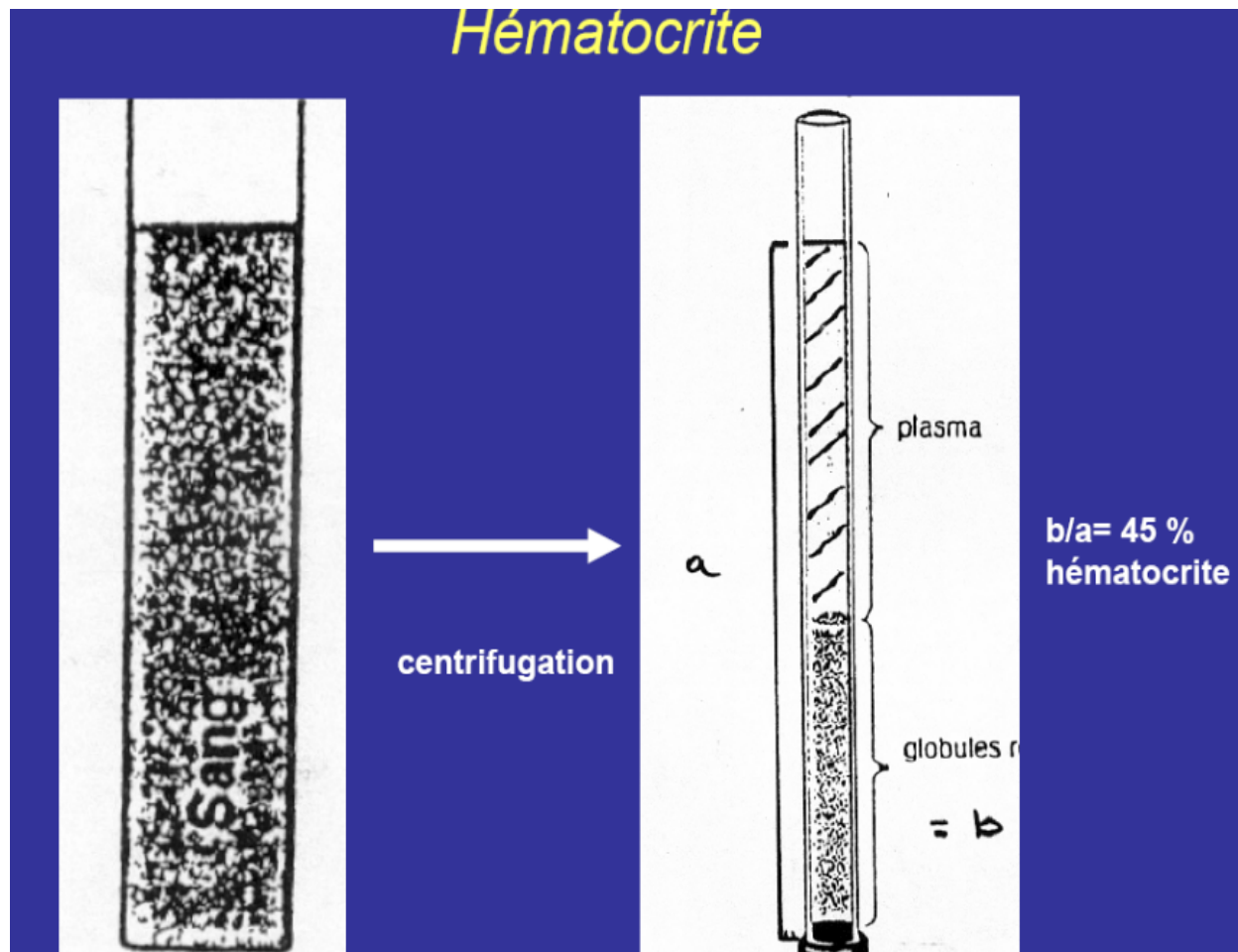


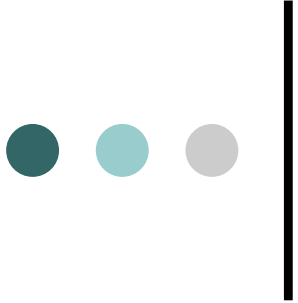
□ Interstitial + Lymphatic volume =

Extracellular volume – Plasma volume

□ 16% of body weight

Hématocrit.

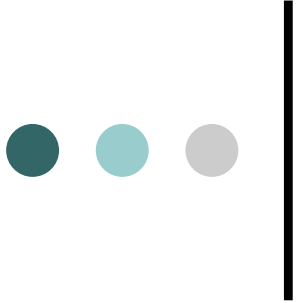




plasma compartment

□ Physical Properties

- Light yellow when fasting
- Lactescent (milky) in postprandial state
- Viscous
- Alkaline, pH 7.35–7.45
- Coagulable



plasma compartment

□ Mineral salts of the plasma compartment:

-Ions

-Blood ionogram

-Electrically neutral

-Na Cl +++

-Osmolality= 300 mosm/L

-H⁺

-pH +++

Ionic composition of plasma

Cations	meq/l
Sodium Na ⁺	142
Potassium K ⁺	4
Calcium Ca ⁺⁺	5
Magnésium m	2
total	153

Anions	meq/l
Chlore Cl ⁻	103
Bicarbonates HCO ₃ ⁻	26
Phosphates H ₂ PO ₄ ⁻ , HPO ₄ ⁻⁻	1.8
Sulfates SO ₄ ⁻⁻	1
Acides organiques (lactates...)	5
Protéines (alb, 0.7, glob, 0.2)	16

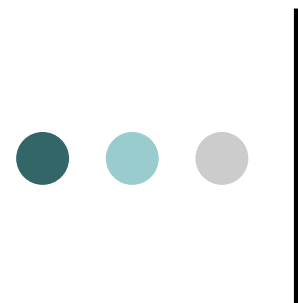
Composition of compartments: electrolytes and dissolved substances

Plasma

Interstitium

Intracellulaire

<u>Cations</u>		
Na ⁺	141	142
K ⁺	4	4
Ca ⁺⁺	5	3,5
Mg ⁺⁺	1,5	1
<u>Anions</u>		
Cl ⁻	103	116
HCO ₃ ⁻	26	29
HPO ₄ ⁻⁻	2	2
SO ₄ ⁻⁻	1	1
ac Org	5	5
Prot-	16	55



Plasma compartment

Plasma pH

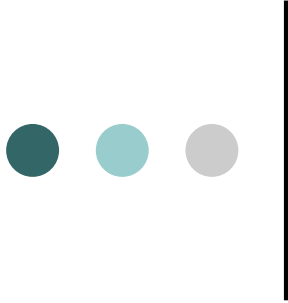
pH = 7.40

Acidosis: pH < 7.35

Alkalosis: pH > 7.45

Cellular activity H⁺

Homeostasis ++



Plasma compartment

□ Plasma pH regulation

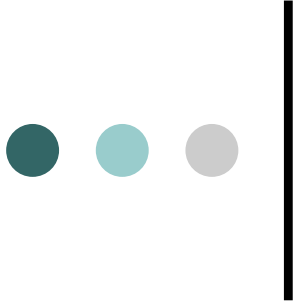
Immediate Regulation => Buffer

Systems $\text{H}^+ + \text{HCO}_3^- \rightleftharpoons \text{H}_2\text{CO}_3$

Bicarbonate / Carbonic acid

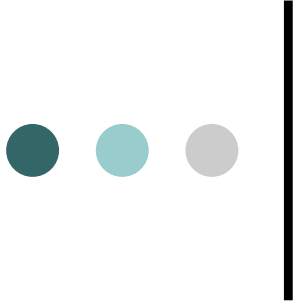
Proteinate / Protein

Disodium phosphate / Monosodium
phosphate



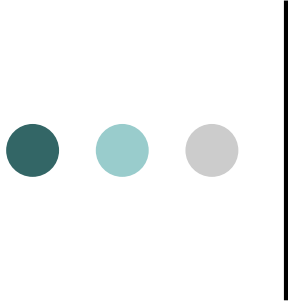
Plasma compartment

- Regulation of Plasma pH
- Immediate regulation = buffer systems
- Delayed regulation
 - Direct = kidneys (H^+)
 - Indirect = lungs (CO_2)



plasma compartment

- Organic protein substances
- Proteins: 70g/L
- Properties:
- Hepatic origin
- Osmotic pressure of the interstitial medium
- Anions – electrical balance
- Buffering capacity



plasma compartment

Albumin (60%)

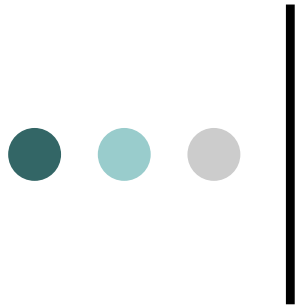
- MW = 65 kDa, composed of approx. 580 AA
 - Osmotic pressure ++
 - Transporters
 - Liver disease: decrease
 - Kidney disease: proteinuria

Globulins (35%)

- MW: 200,000–400,000
- Three groups: \alpha, \beta \gamma
 - gamma globulins:
 - Immunoglobulins ++
 - A, D, E, M, G (IgA, IgD, IgE, IgM, IgG)

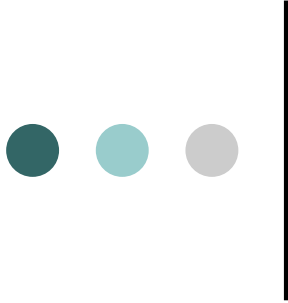
Coagulation and fibrinolysis factors (5%)

- Fibrinogen



plasma compartment

- **Non-Protein Organic Substances**
- **Urea 0.3 g/L**
 - Hepatic origin
 - Urinary elimination
- **Uric acid 0.05 g/L**
 - "Gout"
- **Glucose** → glycemia = 1 g/L
 - Diabetes
 - Hypoglycemia
- **Lipids 5-8 g/L**
 - Triglycerides
 - HDL and LDL Cholesterol



Osmolarity of the extracellular compartment

Concentration en mol/vol plasmatique : mmol/l

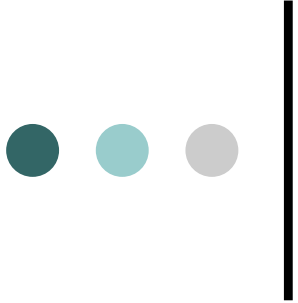
$(Na^+ + K^+) \times 2 + (urée + glucose)$

$(Na^+ + K^+ + Ca^{++} + Mg^{++}) + (Cl^- + HCO_3^- + HPO_4^- \text{ etc.}) + (urée + glucose)$

$\cong 300 \text{ mosmol/l}$

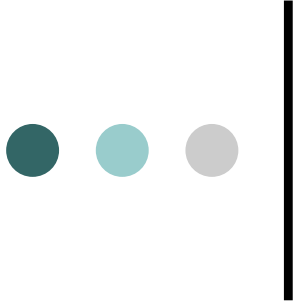
Natrémie = bon reflet de l'osmolarité

Sodium = main cation of the extracellular compartment



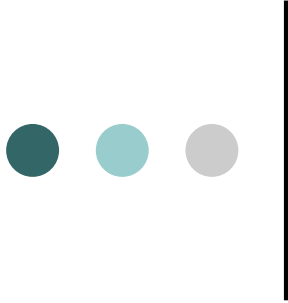
Physicochemical properties of the interstitial compartment

- ❑ **Interstitial Fluid**
- ❑ Does not circulate
- ❑ Exchanges with plasma and cells → **renewal**
- ❑ Cleared by the **lymphatic circulation**
- ❑ Ultrafiltrate of plasma
- ❑ Electrically neutral
- ❑ Proteins: **2 g/L**



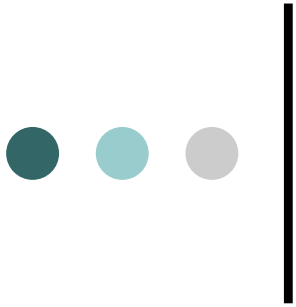
Intracellular compartment

- Total Volume – Extracellular Volume
40% body weight Tissue
inhomogeneity



Physicochemical properties of the intracellular compartment

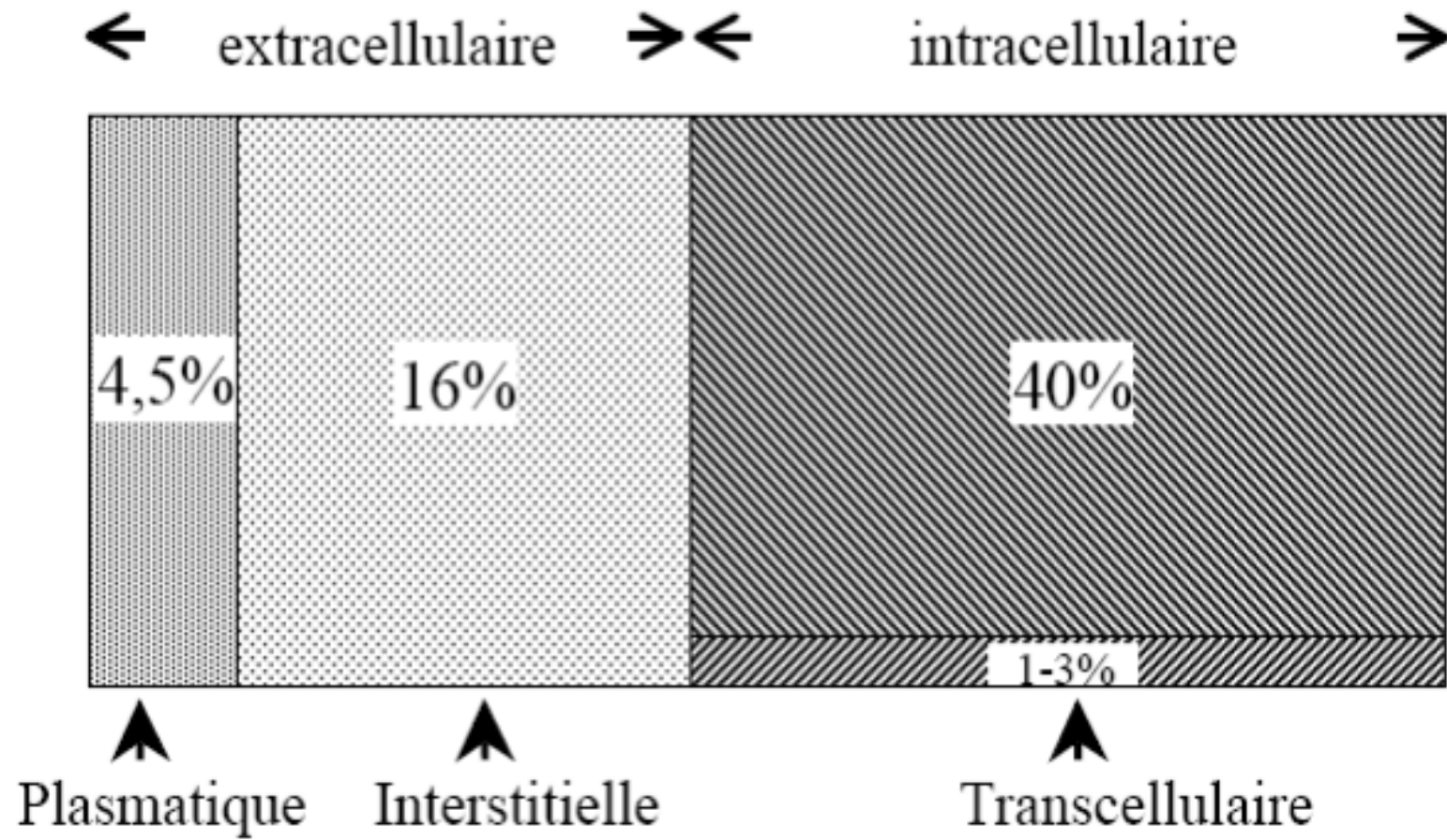
- Intracellular Fluid
- Difficult to analyze
- Cellular heterogeneity
- Osmolarity close to extracellular fluid
- K^+ main cation
- Mg^{2+}

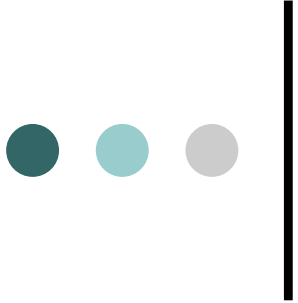


Transcellular compartments

- **Transcellular Fluids**
 - Articular fluid
- Cerebrospinal fluid
 - Digestive fluid
 - Ocular fluid
 - Inner ear fluid
- Glandular fluid

Plasma-interstitium exchanges

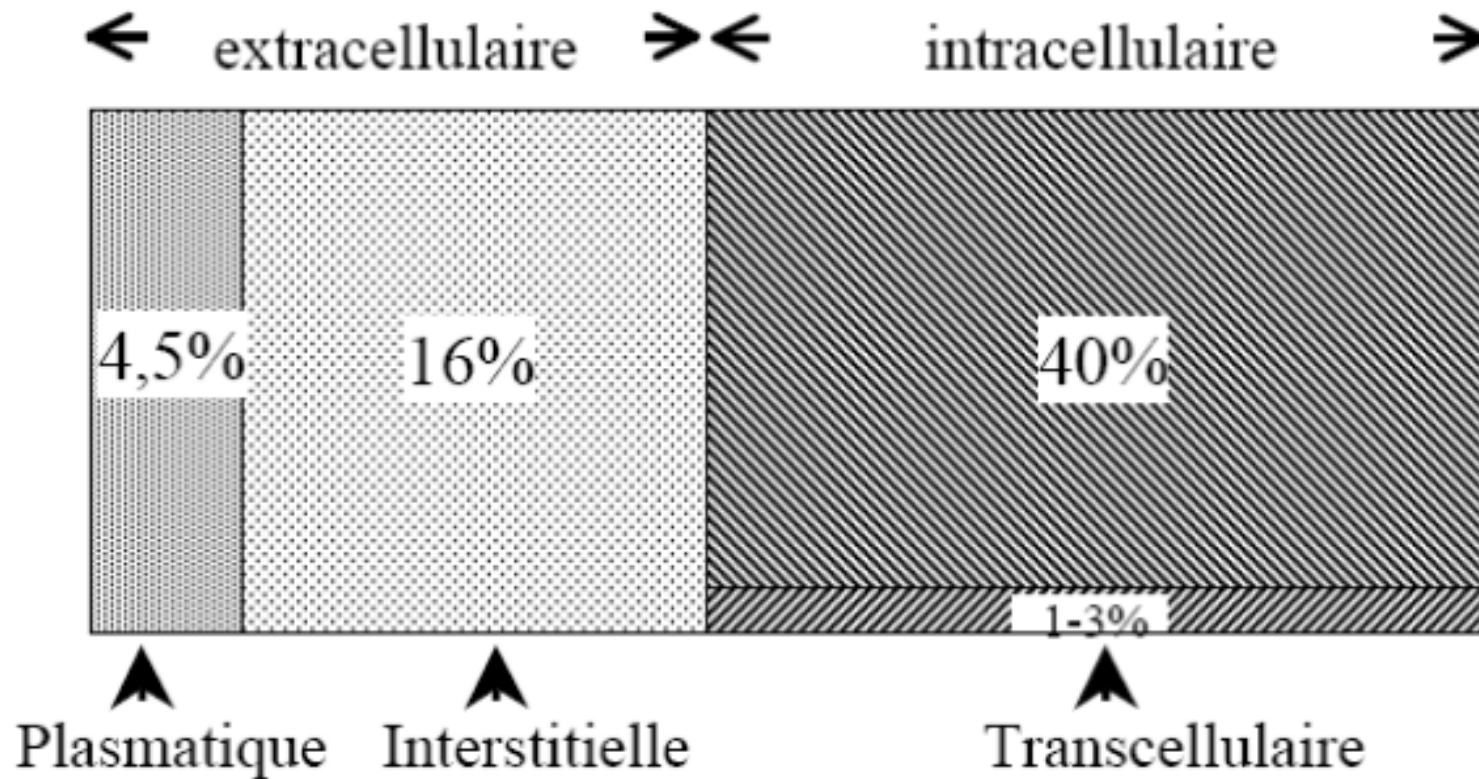


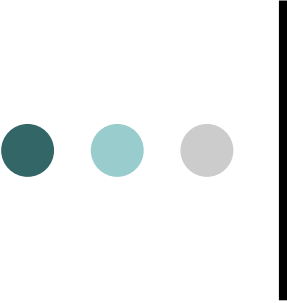


Plasma–interstitium exchanges

- **Extracellular Compartment:** Plasma $\neq$ interstitium (proteins) Interstitial fluid = ultrafiltrate of plasma (protein-free) $\rightarrow$ different anion distribution across the membrane according to the **Gibbs-Donnan equilibrium**
- **Intracellular Compartment:** Intracellular composition $\neq$ extracellular composition

Plasma-interstitium exchanges





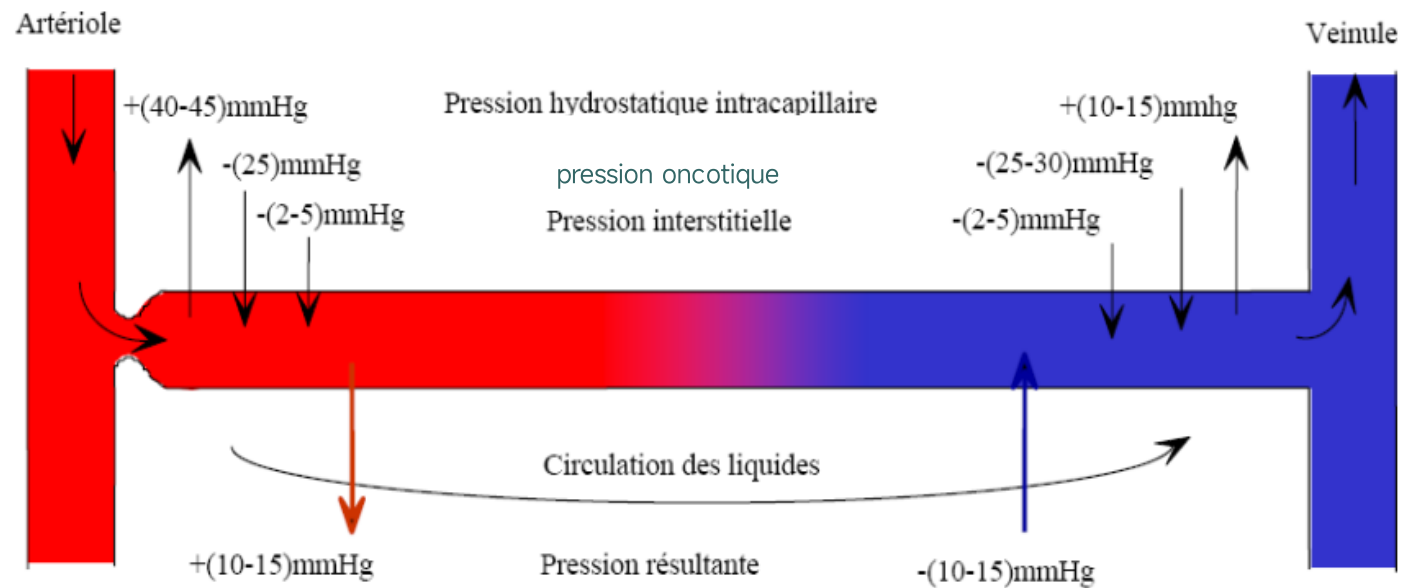
Plasma–interstitium exchanges

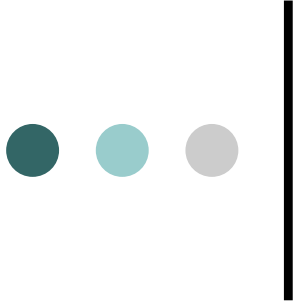
- Water fluxes between these compartments are passive. They depend on osmotic pressure and hydrostatic pressure gradients.
- Vascular wall: a semi-permeable membrane regarding plasma proteins.
- Interstitial fluid = plasma ultrafiltrate (devoid of proteins). Proteins responsible for an osmotic pressure = oncotic pressure.

Plasma-interstitium exchanges

Filtration eau et électrolytes

Réabsorption eau et électrolytes





FLUID COMPARTMENTS

- **Composition of Compartments**
- **Units used:**
- **The mole: M [] molar: MW g/l = 1 M/l**
 - Example Sodium Na^+ : MW = 23
 - 23 g/l = 1 mole/l = 1 molar
 - 23 mg/l = 1 mmole/l
- **The osmole:** for a dissolved substance, concentration determines osmotic pressure
 - 1 mmole/l = 1 mosmole/l = 17 mmHg
- **The equivalent: Eq**, allows the study of charge distribution:
 $\Sigma \text{anions} = \Sigma \text{cations}$